contrary, cooling results in contraction. This increases the density and thus leads to high pressure.

Formation of equatorial low and polar highs are examples of thermal lows and thermal highs, respectively.

Dynamic Factors: Apart from variations of temperature, the formation of pressure belts may be explained by dynamic controls arising out of pressure gradient forces and rotation of the earth (Coriolis force).

12. **Permanent winds or Primary winds or Prevailing winds or Planetary winds**: The trade winds, westerlies and easterlies.

    Secondary or Periodic Winds:
    - Seasonal winds: These winds change their direction in different seasons. For example monsoons in India.
    - Periodic winds: Land and sea breeze, mountain and valley breeze.

    Local winds: These blow only during a particular period of the day or year in a small area. Winds like Loo, Mistral, Foehn, Bora.

13. **Statement 1: CORRECT**—A frontal inversion occurs when a cold air mass undercuts a warm air mass (Cold and Warm Fronts) and lifts it aloft, the front between the two air masses then has warm air above and cold air below.

    Statement 2: INCORRECT—This types of inversion is unstable and is destroyed as the weather changes.

14. Jet streams are:
    - Circumpolar
    - Narrow
    - Concentrated bands of meandering
    - Upper tropospheric
    - High velocity
    - Geostrophic streams
    - Bounded by low speed winds and
    - Part of upper level westerlies

15. **Statement 1: CORRECT**—STJ is closely connected to the Indian and African summer monsoons. Statement 2: INCORRECT—Polar jet stream greatly influences climates of regions lying close to 60 degree latitude.

    Statement 3: CORRECT—Somali jet stream can strengthens or